

OBJECT DETECTION AND TRACKING SYSTEM FOR LOW LIGHT ENVIRONMENT

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OBJECT DETECTION AND TRACKING SYSTEM FOR LOW LIGHT ENVIRONMENT

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by

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DECLARATION

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CERTIFICATE

This is to certify that the Dissertation Report entitled, **OBJECT DETECTION AND TRACKING SYSTEM FOR LOW LIGHT ENVIRONMENT** submitted by **VAD-LAMOODI VENKATA SUBA MAHANTH** to Indian Institute of Technology, Kharagpur, India is a record of bonafide Project work carried out by him/her under my/our supervision and guidance and is worthy of consideration for the award of the degree of Master of Technology in Electrical Engineering of the Institute.

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Contents

List of Figures	vii
List of Tables	viii
Abstract	1
1 Introduction	2
1.1 Background	2
1.2 Motivation	3
1.3 Objective and System Overview	3
1.4 Chapter Overview	4
2 Object Detection	5
2.1 YOLO Overview	5
2.1.1 Backbone	5
2.1.2 Neck	6
2.1.3 Head	6
2.1.4 Why YOLO fails in low-light environments?	6
2.2 Literature Review on Enhancement Methods	7
2.2.1 Visual Quality-Driven Enhancement	7
2.2.2 Recognition Quality-Driven Enhancement	8
2.3 Related Work	8
2.3.1 Why YOLOv8 + PENet for This Thesis	10
3 PENet - YOLO : Model Architecture	11
3.1 PENet Architecture	11
3.1.1 Laplacian Pyramid Decomposition	12
3.1.2 Detail Processing Module (DPM)	13
3.1.3 Low-Frequency Enhancement Filter (LEF)	14
3.1.4 Reconstruction Process	14

3.2	Integration of PENet Into the YOLOv8 Detection Framework	15
3.2.1	Modification of the YOLOv8 Model Configuration	15
3.2.2	Model Construction and Weight Loading	16
3.2.3	Advantages of Simple YAML-Level Integration	16
4	Dataset Overview and Preprocessing	17
4.1	ExDark Dataset	17
4.2	Pre-Processing of ExDark	18
5	Training and Experiments	19
5.1	Training Strategy	19
5.1.1	Pretrained Weight Alignment	19
5.1.2	Stage-1: PENet Warm-Up Training	19
5.1.3	Stage-2: Joint End-to-End Training	20
5.2	Model Variants and Parameter Scale	20
5.2.1	YOLO Loss Function	21
5.3	Experimental Setup	21
6	Results	22
6.1	Laplacian Pyramid Components Visualization	22
6.1.1	How YOLO sees the Input image from PENet	23
6.2	Evaluation Metrics	24
6.3	Model Evaluation	25
6.3.1	Quantitative Evaluation	25
6.3.2	Qualitative Evaluation	27
6.3.3	Failure Case Analysis	27
6.4	Discussion	29
7	Future Work and Conclusion	30
7.1	Enhancement Module Improvements	30
7.2	Tracker Integration Through ROI Enhancement	30
7.3	End to End Detection–Tracking Pipeline	31
7.4	Conclusion	31

List of Figures

1.1	Overview of the proposed low-light detection and tracking system. . . .	4
2.1	YOLO architecture with Backbone, Neck and Head.	6
2.2	Low Light Enhancement Methods based from the survey	7
2.3	Recognition-driven enhancement pipeline guided by detection loss	8
3.1	PENet Architecture	11
3.2	PENet Module Flow	12
4.1	ExDark Dataset Overview	17
4.2	ExDark Dataset samples with Ground Truth Boxes	17
6.1	Visualization of Laplacian pyramid components used in PENet. L (High Freq) G (Low Freq) Each level contains high-frequency residuals highlighting edges and textures at progressively coarser scales.	22
6.2	Comparison of the original input and the PENet-enhanced image fed into YOLO. The enhanced output exhibits restored edges, clearer object structure allowing YOLO to extract stronger features for detection. . . .	23
6.3	Training metrics of PENet + YOLOv8n	25
6.4	Confusion matrix of predictions of 12 classes	26
6.5	Detection Boxes from combined model on some samples	27
6.6	Failure Case 1: Missed dogs and weak person detection in extremely dark scenes.	28
6.7	Failure Case 2: Cat incorrectly classified as Dog under texture loss. . .	28
6.8	Failure Case 3: Missed small/distant bicycles barely visible in deep shadows.	29

List of Tables

2.1	Comparison of Recent Low-Light Object Detection Methods on ExDark Dataset	9
6.1	Performance comparison between baseline YOLOv8 and PENet-enhanced YOLO models on ExDark. (Metrics shown with placeholder values; will be updated after final training.)	25
6.2	Class-wise detection performance of PENet-enhanced YOLOv8n and YOLOv8m on the ExDark dataset.	26

Abbreviations

YOLO	You Only Look Once
UAV	Unmanned Arial Vehicles
mAP	mean Average Precision
PENet	Pyramid Enhancement Network
IoU	Intersection of Union
ExDark	Exclusively Dark
SGD	Stochastic Gradient Desscent

Abstract

Low-light environments severely degrade the performance of computer vision systems due to poor visibility, sensor noise, and the loss of critical semantic information. Traditional object detectors such as YOLO and R-CNN demonstrate strong accuracy under normal illumination but fail to retain stability in adverse lighting. Enhancing images using classical or human-visual-oriented methods often amplifies noise or disrupts edges. The aim of this work is to design a robust, lightweight, and high-performance object detection and tracking system specifically optimized for low-light conditions using signal-processing-inspired image enhancement techniques. By leveraging multi-resolution frequency decomposition, the system seeks to restore meaningful semantics before detection, enabling reliable performance for applications such as UAV surveillance, night-time monitoring, and anomaly tracking.

To address these challenges, This work integrated a Pyramid Enhancement Network (PENet) with the YOLOv8 detector to create an end-to-end trainable architecture capable of enhancing low-light images without requiring paired supervision. PENet decomposes images into Laplacian pyramid components, performing selective enhancement of high- and low-frequency features through a dual-branch Detail Processing Module and a Low-Frequency Enhancement Filter. Experiments conducted on the ExDark dataset demonstrate consistent improvements in mAP over baseline YOLOv8 models, along with qualitative gains in edge clarity, object structure, and contour stability.

Going forward, the system will be extended into a complete low-light detection–tracking pipeline optimized for UAV applications. This includes integrating ROI-based enhancement for trackers, dynamic Sobel filters, learnable reconstruction for the Laplacian pyramid, and real-time optimization for edge devices. The final goal is to build a deployable AI tool capable of detecting, tracking, and recording objects reliably in extreme low-light environments.

Keywords: Low Light Environment, Object Detection and Tracking, Yolo, Image Enhancement

Introduction

Object detection and tracking are fundamental components of modern computer vision systems. Object detection identifies and localizes objects within a scene, while tracking maintains object identities across consecutive frames. Together, these tasks form the basis of intelligent surveillance, autonomous navigation, traffic monitoring, and a wide range of real-time vision applications. Their reliability, however, depends heavily on the quality of visual information captured from the environment. In real-world scenarios where lighting conditions vary widely, existing models struggle to maintain accuracy and stability, especially in low-light environments.

1.1 Background

Traditional object detection frameworks—including early handcrafted-feature-based methods and modern deep-learning models such as YOLO [1], Faster R-CNN [2], and SSD [3]—perform exceptionally well under normal illumination. These models leverage strong contrasts, clear textures, and stable edge information to extract meaningful features for classification and localization. However, when deployed in low-light conditions, their performance deteriorates significantly. Low-light scenes introduce severe challenges such as noise, weak illumination, low contrast, blurred object boundaries, color distortion, and loss of semantic detail. As a result, feature maps extracted from dark images lack the structural cues required by detectors, leading to missed detections, poor localization, and unstable tracking.

Traditional enhancement methods such as histogram equalization or gamma correction brighten images but often amplify noise and distort important textures. Deep learning–based enhancement techniques improve perceptual quality, yet they are typically optimized for human visual appeal rather than improving detection accuracy. Thus, directly applying such enhancement models before detection does not solve the problem and may even degrade performance. This creates a pressing need for an enhancement strategy that is explicitly designed to support machine perception and improve detection performance under adverse lighting.

1.2 Motivation

Low-light object detection and tracking are critical for several real-time applications:

- **Traffic Monitoring:** Fails under low illumination due to weak edges and noisy feature maps, leading to missed or unstable vehicle detections.
Improvement: Illumination-aware enhancement can restore contrast and strengthen motion cues for reliable anomaly detection.
- **Border Surveillance:** Night-time scenes cause detectors to lose semantic detail, reducing the ability to identify suspicious activity.
Improvement: Task-oriented enhancement can recover structure and suppress noise to support high-confidence detection.
- **UAV-based Aerial Surveys:** Varying altitude and dim lighting degrade visual quality, causing tracking drift or complete ID loss.
Improvement: Lightweight enhancement modules help maintain discriminative features and stable tracking across frames.

These limitations motivate an integrated enhancement–detection and tracking framework that reinforces semantic cues, improves feature quality, and delivers robust performance in low-light conditions.

1.3 Objective and System Overview

The objective of this work is to develop a low-light object detection and tracking system by integrating an image enhancement module with Image Processing Techniques and Deep Learning. The complete pipeline has a Detector, Tracker and Enhancement Module. For a Real time system or on the Input Video System, Frames will undergo Detection and unique IDs will be generated when its required to Track/Trace particular object. If tracking is lost then Detection will be re-initiated and it will be a loop. For Detections, overall Image Enhancement is must but for Tracking **DarkLighter** [4] Proposes that global enhancement is often unnecessary for tracking; instead, region-of-interest (ROI) enhancement can selectively improve object appearance without amplifying background noise. This motivates the integration of lightweight enhancement modules into the tracking pipeline, enabling trackers to maintain stronger discriminative features under low-light conditions. We can automate this triggering tracker, in case of a classification system that needs to identify any suspicious activity among the detected objects.

As illustrated in Figure 1.1, the system begins with a video or image input captured under low-light conditions. The enhancement module processes each frame to recover

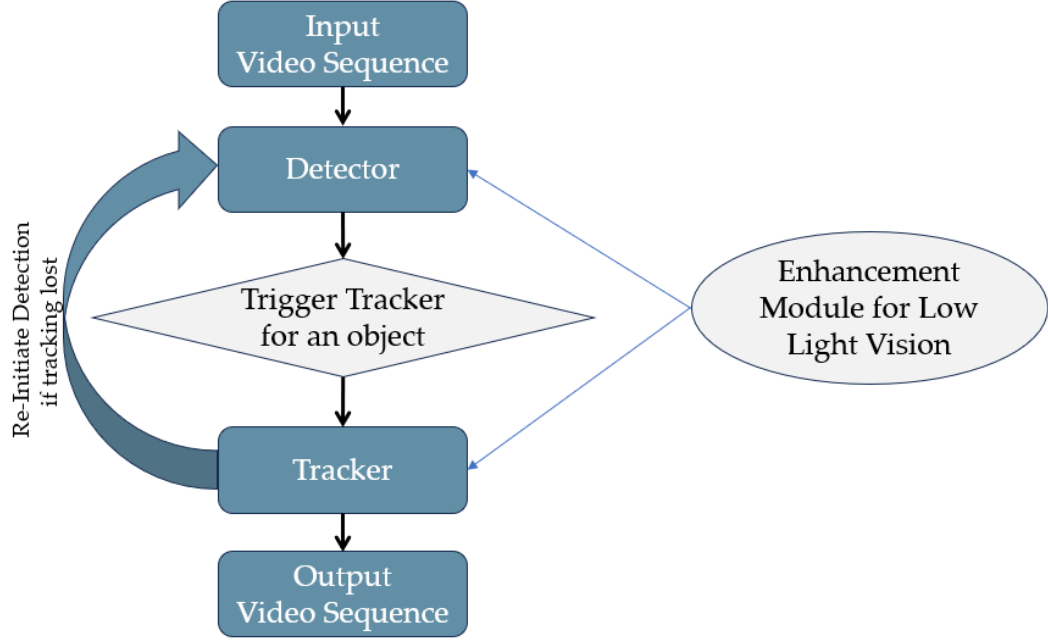


Figure 1.1: Overview of the proposed low-light detection and tracking system.

structural details and improve contrast. The enhanced output is then passed to a YOLO-based detector for generating bounding boxes and class predictions. Finally, a tracking module such as DeepSORT or SORT links detections across time to maintain persistent object identities. This modular yet unified pipeline forms the foundation for developing a reliable, real-time intelligent vision system for adverse lighting environments.

1.4 Chapter Overview

Chapter 2: Object Detection and Literature Review on Low light Enhancement Methods

Chapter 3: Model Architecture, Enhancement module, and integration with detectors

Chapter 4: Description of the dataset, Dataset Preprocessing.

Chapter 5: Training, Experimental results and performance analysis.

Chapter 6: Discussion of future work and conclusion of the thesis.

Object Detection

Object detection is a fundamental stage of our system, forming the basis for identifying and localizing targets before tracking and further analysis can occur. This chapter focuses specifically on the evolution of modern detection models, with emphasis on real-time architectures such as the YOLO family and their relevance to our pipeline. We outline how these detectors extract hierarchical features, how their design enables fast inference, and why their performance deteriorates under low-light conditions. The chapter also reviews existing low-light enhancement approaches and how they influence detection reliability. By examining both detection models and enhancement-driven improvements, this chapter establishes the technical foundation for the enhancement-augmented detection framework developed in this thesis.

2.1 YOLO Overview

The YOLO (**You Only Look Once**) family represents one of the most influential real-time object detection frameworks, designed to perform classification and localization in a single forward pass. Unlike two-stage detectors such as Faster R-CNN, which generate region proposals before classification, YOLO adopts a fully convolutional single-stage design that enables high inference speed while maintaining competitive accuracy. The architecture is broadly composed of three functional components: the *Backbone*, *Neck*, and *Head*, each responsible for different stages of feature extraction and prediction.

2.1.1 Backbone

The backbone extracts feature representations from the input image using stacked convolutional and residual/CSP blocks. It produces multi-scale feature maps containing low-level edges, mid-level textures, and high-level semantic cues that form the foundation of the detector.

2.1.2 Neck

The neck fuses feature maps from different backbone stages to improve detection across object scales. Using structures such as FPN or PAN, it combines shallow spatial details with deeper semantic features, ensuring both small and large objects are effectively represented.

2.1.3 Head

The head generates the final predictions, including bounding boxes, objectness scores, and class probabilities. Modern YOLO heads separate classification and regression branches and output predictions at multiple scales. Non-Maximum Suppression (NMS) is applied to remove overlapping detections.

Overall, the YOLO architecture achieves a strong balance between accuracy and real-time performance through efficient feature extraction, multi-scale feature fusion, and streamlined prediction heads. These characteristics make YOLO an ideal backbone detector for integration with low-light enhancement modules in the system developed in this thesis.

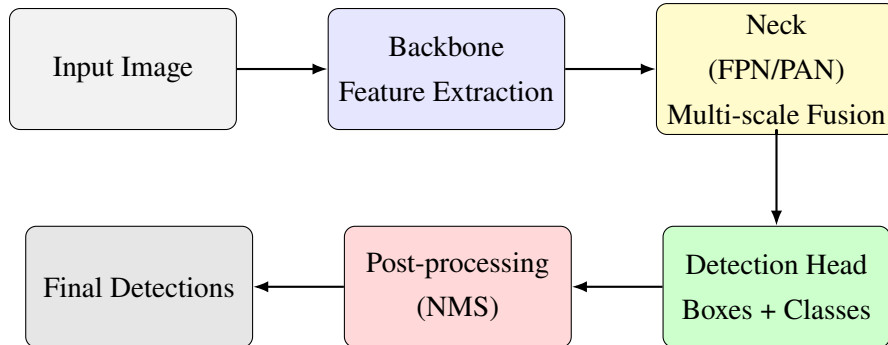


Figure 2.1: YOLO architecture with Backbone, Neck and Head.

2.1.4 Why YOLO fails in low-light environments?

YOLO detectors rely on strong gradients, clear edges, and stable textures to build hierarchical feature maps in the early convolutional layers. In low-light conditions, these cues weaken due to reduced signal-to-noise ratio (SNR), noise, and loss of contrast. As illumination drops, high-frequency details diminish, causing weak activations in the backbone’s initial filters. This loss of early structural cues propagates through deeper layers, leading to poor semantic feature representation.

Architecturally, YOLO’s CSP and C2f blocks depend on residual aggregation of meaningful features; when inputs are noise-dominated, these paths amplify noise and destabilize feature maps. The PAN/FPN neck further mixes low-contrast features across scales, reducing small-object discriminability. Since the detection head relies on

consistent spatial gradients for bounding-box regression and classification, suppressed gradients in dark regions produce low-confidence, mislocalized, or completely missed detections. Thus, YOLO fails in low light because informative features collapse across the entire pipeline—from early convolutions to multi-scale fusion and final prediction.

2.2 Literature Review on Enhancement Methods

Low-light image enhancement (LLIE) has evolved significantly with the growth of deep learning, and recent literature categorizes enhancement techniques into two major paradigms: *visual quality-driven* and *recognition quality-driven* methods [5]. This distinction is central to understanding how enhancement models influence high-level tasks such as object detection and tracking.

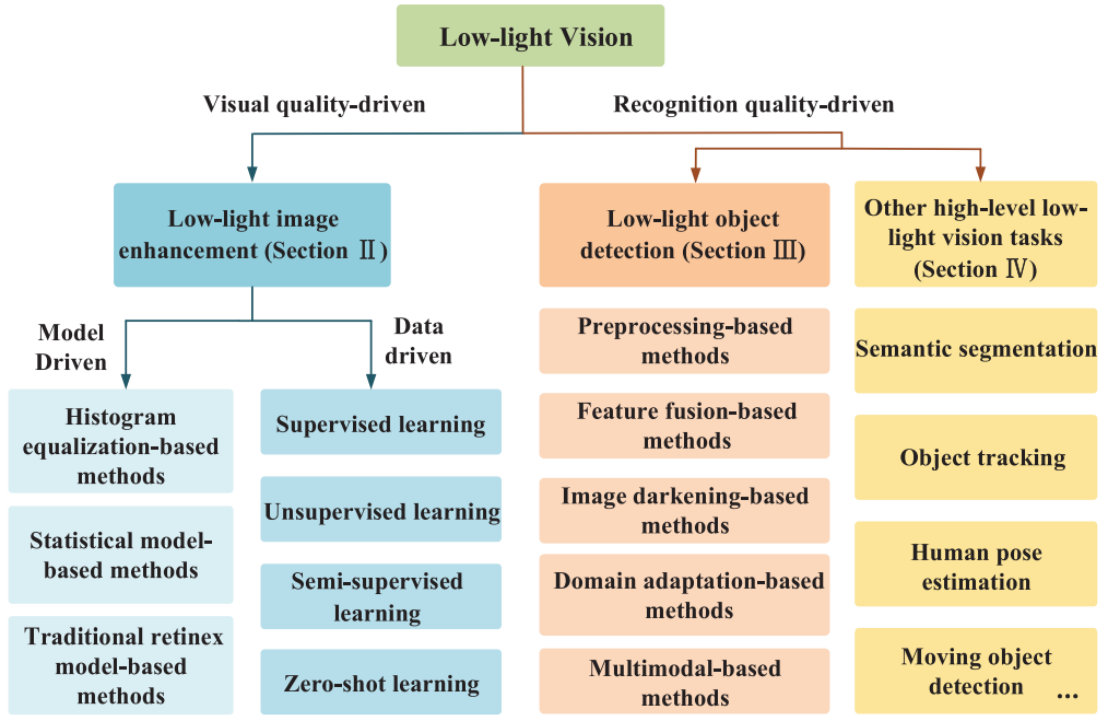


Figure 2.2: Low Light Enhancement Methods based from the survey

2.2.1 Visual Quality-Driven Enhancement

Visual-quality LLIE methods focus on improving the perceptual appearance of dark images by enhancing brightness, contrast, and color fidelity. Traditional approaches such as histogram equalization and Retinex-based correction improve visibility but often amplify noise under uneven illumination. Modern deep-learning models (e.g., Retinex-Net, KinD, URetinexNet, EnlightenGAN, Zero-DCE) generate visually pleasing results using learned illumination priors and pixel-level metrics such as PSNR, SSIM,

NIQE, and LPIPS. However, these methods are optimized for human perception and may not preserve the semantic features required for reliable object detection.

2.2.2 Recognition Quality-Driven Enhancement

Recognition-driven LLIE approaches enhance images specifically to benefit downstream tasks such as object detection and tracking. These methods prioritize retaining discriminative features, edge clarity, and noise robustness rather than visual aesthetics. Recent frameworks jointly train the enhancement and detection modules or operate in the feature domain to avoid pixel-level artifacts. Such task-oriented approaches consistently outperform visual-quality methods when integrated with detectors like YOLO or transformer-based models.

Overall, the survey shows that **visually enhanced images don't necessarily improve detection performance**. Effective low-light detection requires enhancement aligned with recognition semantics, which motivates our task-driven PENet-based enhancement.

2.3 Related Work

Recent research has explored enhancement-driven object detection to address the challenges posed by low-light environments. Works such as featEnHancer [6] and LightStar [7] focus on improving feature quality prior to detection, demonstrating the benefits of integrating enhancement into the detection pipeline. According to the taxonomy presented in the comprehensive survey on low-light vision [5], these approaches fall under **recognition-quality driven** enhancement, where the objective is not visual aesthetics but improved downstream performance. In such systems, enhancement modules are trained jointly with detectors such as YOLO, enabling the enhancement network to preserve semantic structure, amplify useful contrast cues, and suppress illumination noise.

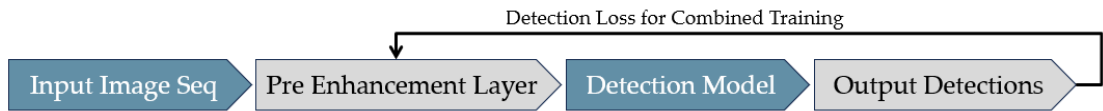


Figure 2.3: Recognition-driven enhancement pipeline guided by detection loss

A wide range of recognition-oriented enhancement–detection frameworks have emerged, including domain adaptation models (DAI-Net [8]), pseudo-RAW based enhancement (LightStar-YOLO [7]), feature-focused enhancement networks (DBLD-Net [9], MEEM-YOLO [10]), attention-based approaches (En-YOLO [11], CBAM-YOLO), and RAW-level processing pipelines such as Dark-ISP [12]. Table 2.1 provides comparison of recent low-light detection methods evaluated on the ExDark benchmark.

Table 2.1: Comparison of Recent Low-Light Object Detection Methods on ExDark Dataset

Model	Method	mAP@50%	Year	Publication
PE-YOLO	Pyramid Enhancement Network for Dark Objects [13]	78.0	2023	arXiv (ICANN)
DAI-Net	Detection with zero-shot day–night domain adaptation [8]	78.3	2024	arXiv (CVPR)
LightStar-YOLO	Pseudo-RAW Space Enhancement LLIE method [7]	78.6	2025	Springer (IC-CVM)
YOLO-AS	Zero-DCES enhancement + attention (ECA_ASPP) [14]	78.39	2025	Scientific Reports
Dark-ISP	RAW-level ISP correction [12]	70.4 (ResNet50)	2025	arXiv (ICCV)
DBLD-Net	Dual-branch multi-scale enhancement [9]	76.1	2025	arXiv
Dark-DeNet	Lightweight enhancement module [15]	69.4 (YOLOv5)	2025	IEEE
En-Yolo	Dual-attention feature learning [11]	73.46	2025	Springer
MEEM (YOLO)	Multi-scale edge enhancement [10]	75.9	2025	Springer
CBAM-YOLO	CBAM attention applied to YOLOv11	74.1	2025	IEEE

Among these methods, **PE-YOLO** stands out as particularly relevant. PE-YOLO introduces the Pyramid Enhancement Network (PENet) [13], which uses Laplacian pyramid decomposition and multi-resolution enhancement to restore structure and illumination before joint training with the detector. The network is lightweight (90k parameters), computationally efficient, and consistently improves mAP across several dark-object categories on ExDark. Given its balance of simplicity, effectiveness, and compatibility with modern detectors, PENet from PE-YOLO is adopted in this thesis as the core enhancement module integrated with YOLOv8.

2.3.1 Why YOLOv8 + PNet for This Thesis

YOLOv8 is chosen for its strong baseline performance, robust feature extraction, and efficient backbone–neck–head design. PNet complements YOLOv8 by enhancing multi-frequency information prior to feature extraction. This combination offers several advantages:

- **High compatibility:** PNet can be inserted as a plug-in enhancement layer at the start of the YOLOv8 backbone without architectural conflicts.
- **Lightweight enhancement:** Only ~90k parameters are added, maintaining real-time performance.
- **Multi-resolution restoration:** The Laplacian pyramid enhances both low-frequency illumination and high-frequency edges, improving feature consistency for YOLOv8.
- **Task-driven improvement:** Following PE-YOLO’s results [13], PNet enhances structural information aligned with detection loss rather than visual quality.

Thus, YOLOv8 combined with PNet provides a solid foundation for building an enhancement-driven detection system, which is further extended in this work toward a complete detection–tracking pipeline for low-light scenarios.

PENet - YOLO : Model Architecture

The previous chapter reviewed existing object detection frameworks and low-light enhancement methods, highlighting the limitations of conventional detectors in dark environments and the need for task-oriented enhancement. Building upon these insights, this chapter introduces the enhancement-integrated detection framework (PENet+YOLO) designed for low-light object detection. We describe the overall system architecture, the design of the Pyramid Enhancement Network (PENet), and its integration with the YOLO-based detection pipeline.

3.1 PENet Architecture

The block diagram of the Pyramid Enhancement Network (PENet) is shown in Figure 3.1. This section explains about the Enhancement and its integration with Detection model.

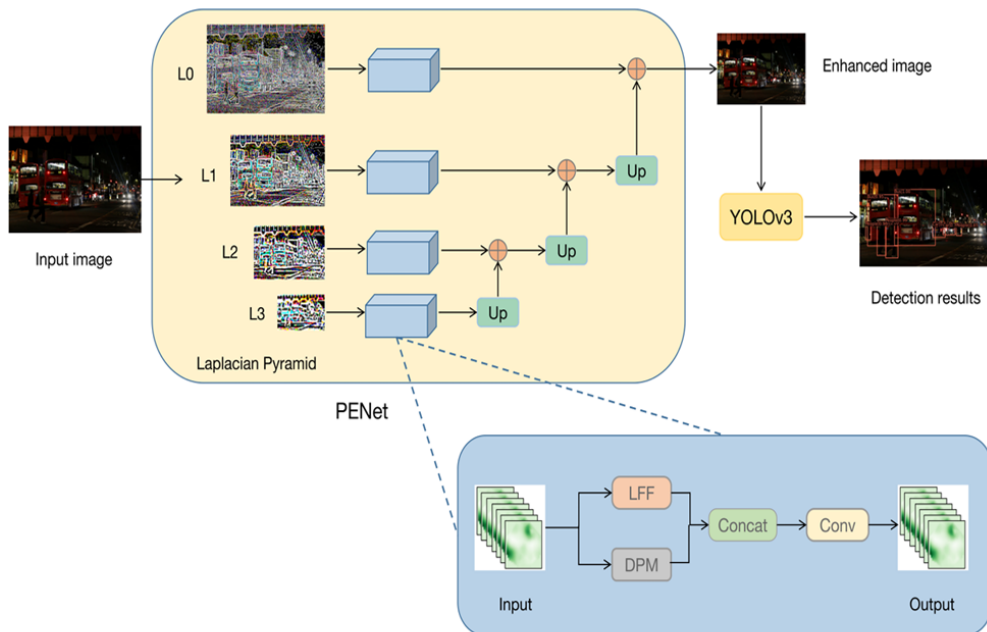


Figure 3.1: PENet Architecture

To improve robustness in low-light conditions, the proposed system adopts the Pyramid Enhancement Network (PENet), originally introduced in PE-YOLO [13]. PENet enhances an image by decomposing it into multiple frequency components, processing each component with specialized modules, and reconstructing an enhanced representation suited for object detection. The decomposition into four pyramid scales enables the network to separately learn global low-frequency illumination information and high-frequency texture details, which is essential for low-light scenes where brightness, contrast, and edges deteriorate at different rates. This section describes each stage of the enhancement pipeline.

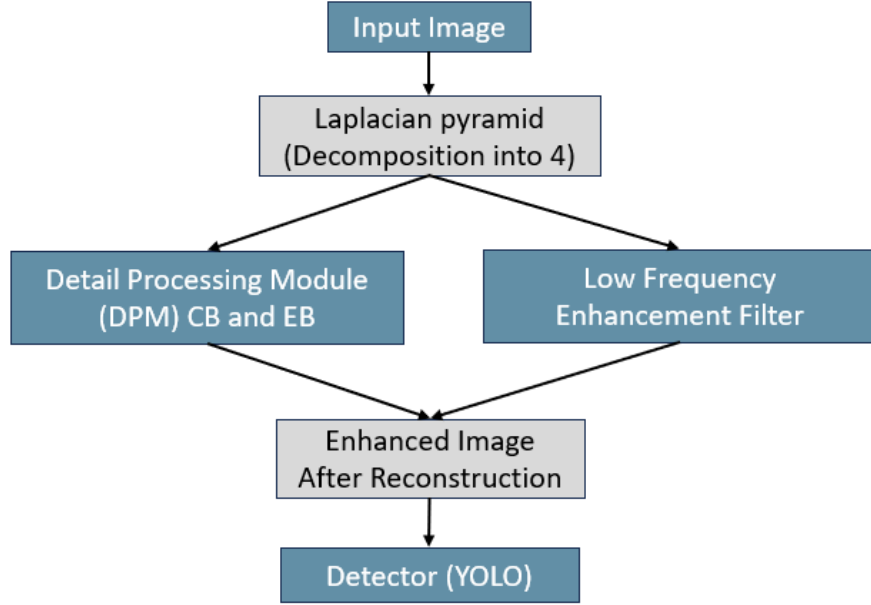


Figure 3.2: PENet Module Flow

3.1.1 Laplacian Pyramid Decomposition

Given an input image $I \in \mathbb{R}^{h \times w \times 3}$, the Laplacian pyramid decomposes it into multi-resolution components capturing both low-frequency illumination structure and high-frequency textures. A Gaussian pyramid is first constructed:

$$G_0 = I, \quad G_1 = \text{Down}(G_0), \quad G_2 = \text{Down}(G_1), \quad G_3 = \text{Down}(G_2) \quad (3.1)$$

where each downsampling reduces the spatial resolution by a factor of 2, producing feature maps of size:

$$h \times w, \quad \frac{h}{2} \times \frac{w}{2}, \quad \frac{h}{4} \times \frac{w}{4}, \quad \frac{h}{8} \times \frac{w}{8}. \quad (3.2)$$

These represent the low-frequency components at different scales.

The corresponding high-frequency Laplacian components are obtained by subtracting the upsampled coarse level from the finer level:

$$L_0 = G_0 - \text{Up}(G_1), \quad L_1 = G_1 - \text{Up}(G_2), \quad L_2 = G_2 - \text{Up}(G_3), \quad L_3 = G_3. \quad (3.3)$$

Thus, the decomposition yields four levels:

- **Low-frequency maps:** G_0, G_1, G_2, G_3 (illumination, smooth structure)
- **High-frequency maps:** L_0, L_1, L_2, L_3 (edges, textures, fine details)

The benefit of this decomposition is that *illumination degradation* mainly affects low-frequency bands, while noise and texture loss affect high-frequency bands. By enhancing them separately, PENet provides fine-grained control over luminance and detail recovery.

3.1.2 Detail Processing Module (DPM)

The high-frequency components $\{L_i\}$ contain local textures and fine structural information that are often lost in low-light environments. To enhance these details, the Detail Processing Module (DPM) processes each high-frequency map through two parallel branches:

1. Context Branch (CB)

This branch captures global contextual information by modeling long-range dependencies. Given a component x , the context-enhanced representation is:

$$\hat{x} = \sigma(F_2(x)) \cdot x \quad (3.4)$$

$$CB(x) = x + \gamma(F_1(\hat{x})) \quad (3.5)$$

where F_1 and F_2 are 3×3 convolutions, σ is a softmax-based attention, and γ is a LeakyReLU activation. This enhances large-scale structural cues such as object boundaries and smooth transitions.

2. Edge Branch (EB)

The edge branch focuses on recovering missing edges by applying Sobel filters:

$$EB(x) = F_3(\text{Sobel}_h(x) + \text{Sobel}_v(x)) + x \quad (3.6)$$

where horizontal and vertical gradient responses are recombined and refined using a convolutional layer. This restores sharp details and improves texture visibility for detectors like YOLO, which rely heavily on edges.

The outputs of CB and EB are fused to obtain the enhanced high-frequency map:

$$L'_i = CB(L_i) + EB(L_i). \quad (3.7)$$

3.1.3 Low-Frequency Enhancement Filter (LEF)

Low-frequency components $\{G_i\}$ represent global illumination and semantic structure. In low-light images, these components suffer from poor visibility and uneven brightness. The LEF enhances these components using ****multi-scale adaptive average pooling****, acting as dynamic low-pass filters.

Given a low-frequency map $f \in \mathbb{R}^{h \times w \times 3}$, PENet first projects it to 32 channels and splits it into four groups:

$$f \rightarrow (f_1, f_2, f_3, f_4) \quad (3.8)$$

Each group is processed using a different pooling scale: - 1×1 (global illumination) - 2×2 (coarse spatial smoothing) - 3×3 (mid-level structure) - 6×6 (global semantic cues)

For each:

$$\text{LEF}(f_i) = \text{Up}(\beta_s(f_i)) \quad (3.9)$$

where β_s is adaptive pooling and Up is bilinear upsampling. Finally, all filtered outputs are concatenated and projected back to 3 channels to form the enhanced low-frequency component:

$$G'_i = \text{Concat}(\text{LEF}(f_1), \dots, \text{LEF}(f_4)) \quad (3.10)$$

This selectively enhances illumination while suppressing high-frequency noise, improving detector-friendly global visibility.

3.1.4 Reconstruction Process

After enhancing both low-frequency (G'_i) and high-frequency (L'_i) components, the Laplacian pyramid is reconstructed from the coarsest scale upward:

$$\begin{aligned} G'_3 &\text{ is the coarsest enhanced level,} \\ G'_2 &= L'_2 + \text{Up}(G'_3), \\ G'_1 &= L'_1 + \text{Up}(G'_2), \\ I' &= L'_0 + \text{Up}(G'_1). \end{aligned} \quad (3.11)$$

This produces the final enhanced image I' that is fed into the YOLO detector.

Reconstruction ensures that:

- Low-frequency enhancement improves global illumination

- High-frequency enhancement restores textures and edges
- The combined image remains structurally consistent
- The detector receives a high-quality, noise-suppressed input

This end-to-end differentiable enhancement pipeline allows PENet to learn directly from detection loss, making it effective and lightweight for real-time object detection and tracking.

3.2 Integration of PENet Into the YOLOv8 Detection Framework

To incorporate enhancement directly into the detection pipeline, the PENet module is inserted as an additional layer at the top of the YOLOv8 backbone [16]. This modification ensures that every input image is first processed by the enhancement network before feature extraction begins. Unlike external pre-processing, this design makes PENet a structural component of the detector, allowing end-to-end training driven entirely by the YOLO loss.

3.2.1 Modification of the YOLOv8 Model Configuration

The YOLOv8 architecture is defined using a YAML configuration file (`yolov8.yaml`). In the original configuration, the backbone begins directly with a convolutional layer. To integrate PENet, we simply prepend a new module entry in the backbone list:

```
# Original YOLOv8 backbone:
backbone:
  - [Conv, 3, 32, 3, 2]
  - [C2f, 32, 64, 3]
  ...

# Modified backbone with PENet:
backbone:
  - [PENet]          # <-- Inserted as first layer
  - [Conv, 3, 32, 3, 2]
  - [C2f, 32, 64, 3]
  ...
```

No additional parameters are required inside the bracket; the corresponding PENet class defined in the model source file handles all internal operations. Thus, YOLOv8

automatically treats PENet as a valid module and routes the input image through it before passing it to the backbone layers.

3.2.2 Model Construction and Weight Loading

After modifying the YAML, the combined model is instantiated by loading the new configuration. Pretrained YOLOv8 weights are partially loaded—layers that match the original architecture receive pretrained weights, while the newly inserted PENet layers are initialized from scratch. This preserves the strong feature extraction ability of YOLOv8 while allowing PENet to gradually learn enhancement behavior optimized for detection.

The resulting processing pipeline becomes:

$$I \rightarrow \text{PENet} \rightarrow \text{YOLOv8 Backbone} \rightarrow \text{Neck} \rightarrow \text{Detection Head}.$$

3.2.3 Advantages of Simple YAML-Level Integration

- **Minimal architectural changes:** Only a single line is added to the YAML file.
- **Modular and clean:** PENet remains a standalone Python module, making the system easy to maintain.
- **End-to-end optimization:** The enhancement module receives gradients from YOLO loss and adapts to detection.
- **Lightweight insertion:** The real-time performance of YOLOv8 is preserved.

This streamlined integration strategy results in a unified detection architecture where enhancement is tightly coupled with the backbone, enabling improved performance under low-light conditions.

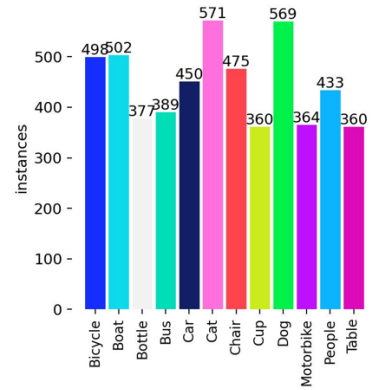
Dataset Overview and Preprocessing

4.1 ExDark Dataset

In this work, we Train and evaluate our enhancement-integrated detection framework on the **ExDark (Exclusively Dark) Dataset** [17], a benchmark specifically created for low-light object detection. ExDark contains real-world night-time images captured under diverse illumination conditions, including dim, ambient, low-light, and extremely dark scenes. The dataset provides object-level annotations and consists of 12 object categories: *bicycle, boat, bottle, bus, car, cat, chair, cup, dog, motorbike, person, and table*.



(a) Sample images from the ExDark Dataset



(b) Distribution of classes

Figure 4.1: ExDark Dataset Overview

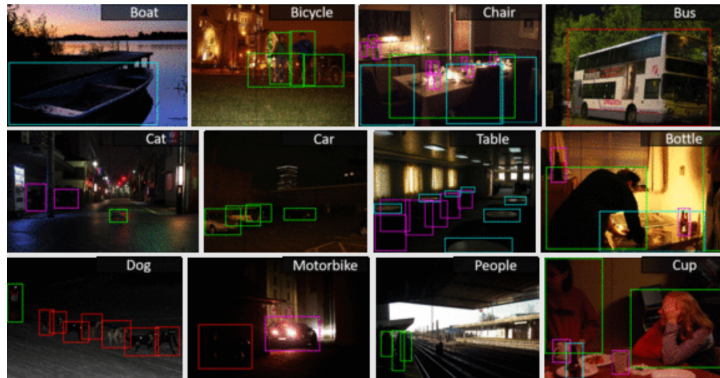


Figure 4.2: ExDark Dataset samples with Ground Truth Boxes

ExDark contains a total of **7361** labeled images. we split dataset into: 80% and 20% for training and validation. Training set consisting 5888 and validation set of 1473 Image and label pairs.

4.2 Pre-Processing of ExDark

The original ExDark dataset provides bounding box annotations in a category-specific text format, where each label includes the class name together with pixel-based bounding box coordinates. To train YOLOv8 on ExDark, the dataset was reorganized into the standard YOLO directory structure and all annotations were converted into the YOLO label format.

Annotation Conversion

Each ExDark annotation was converted into the YOLO label format:

$$\langle \text{class_id} \rangle \langle \text{x_center} \rangle \langle \text{y_center} \rangle \langle \text{width} \rangle \langle \text{height} \rangle$$

where all bounding box coordinates are normalized to the range $[0, 1]$ with respect to the image width and height.

The conversion process consisted of:

1. Reading the original bounding box in pixel format: $(x_{\min}, y_{\min}, x_{\max}, y_{\max})$.
2. Computing normalized YOLO coordinates:

$$\begin{aligned} x_c &= \frac{x_{\min} + x_{\max}}{2W}, & y_c &= \frac{y_{\min} + y_{\max}}{2H}, \\ w &= \frac{x_{\max} - x_{\min}}{W}, & h &= \frac{y_{\max} - y_{\min}}{H}. \end{aligned} \tag{4.1}$$

3. Mapping the object class name to its numeric class index and writing one class per object to a `.txt` file.

Apart from annotation conversion, Other preprocessing steps include:

- **Image Resizing:** All images were resized to 640×640 while preserving aspect ratio using letterboxing. This ensures consistent input dimensions for YOLOv8.
- **Directory Structuring:** The dataset was arranged into the standard YOLO format:

$$\text{datasets/ExDark/\{images, labels\}/\{train, val\}}$$

with identical filenames maintained for image–label pairs.

Training and Experiments

This chapter describes the complete training procedure used to optimize the proposed PENet-enhanced YOLOv8 detector. The addition of the PENet module modifies the original YOLOv8 architecture, making direct loading of pretrained weights infeasible due to the layer mismatch introduced at the first stage of the backbone. To ensure stable training and preserve the benefits of pretrained YOLO features, we adopt a two-stage training strategy consisting of (i) warm-up training with the YOLO backbone frozen, followed by (ii) full end-to-end joint optimization.

5.1 Training Strategy

5.1.1 Pretrained Weight Alignment

In the modified architecture, PENet is inserted as the first layer of the YOLOv8 backbone. Since this shifts all subsequent layers by one position, the pretrained YOLOv8 weights cannot be loaded directly; the model would otherwise fail due to incompatible layer indices. To address this issue, pretrained YOLOv8 weights (fine-tuned on the ExDark dataset) are manually mapped to the corresponding layers of the combined model:

$$\text{Combined_Model}[i + 1] \leftarrow \text{YOLOv8_Pretrained}[i]$$

ensuring that all original YOLOv8 backbone, neck, and head layers retain their pretrained initialization, while the newly introduced PENet block is initialized randomly.

5.1.2 Stage-1: PENet Warm-Up Training

If the combined model were trained directly from scratch, the randomly initialized PENet would disrupt the pretrained feature distribution of YOLOv8, leading to unstable gradients and poor convergence. To avoid this, we freeze the entire YOLOv8 backbone, neck, and detection head during the initial warm-up stage, allowing only the PENet parameters to update. This enables PENet to learn basic illumination correction and frequency enhancement behavior without affecting the pretrained detection layers.

- Trainable Modules: PENet only
- Frozen Modules: All YOLOv8 layers
- Duration: 10 epochs
- Optimizer: SGD
- Learning Rate: 0.001

This stabilizes the enhancement network and prevents catastrophic degradation of YOLO features during early training.

5.1.3 Stage-2: Joint End-to-End Training

After PENet learns initial enhancement behavior, the entire combined model is unfrozen and trained jointly. At this stage, gradients propagate through PENet, the YOLOv8 backbone, PAN-FPN neck, and the detection head, allowing the enhancement module to adjust its output to maximize detection performance.

- Trainable Modules: PENet + YOLOv8
- Duration: 30 epochs
- Optimizer: SGD
- Learning Rate: 0.001

This second-stage optimization aligns the enhancement and detection modules into a unified end-to-end system.

5.2 Model Variants and Parameter Scale

To evaluate the effect of model size, experiments were conducted with two YOLOv8 architectures:

- **YOLOv8n**: 3M parameters + 91k PENet parameters
- **YOLOv8m**: 25M parameters + 91k PENet parameters

The PENet component remains lightweight (approximately 0.09M parameters) and adds minimal computational overhead to both YOLO variants.

5.2.1 YOLO Loss Function

The YOLO loss combines three components: bounding box regression loss, objectness loss, and classification loss. For each prediction, the total loss is

$$\mathcal{L}_{\text{YOLO}} = \lambda_{\text{box}} \mathcal{L}_{\text{box}} + \lambda_{\text{obj}} \mathcal{L}_{\text{obj}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}. \quad (5.1)$$

1) Bounding Box Regression Loss YOLO uses IoU-based losses (e.g., CIoU/DIoU/GIoU) to measure the quality of predicted bounding boxes:

$$\mathcal{L}_{\text{box}} = 1 - \text{IoU}(b_{\text{pred}}, b_{\text{gt}}). \quad (5.2)$$

2) Objectness Loss Objectness reflects whether an object exists in a predicted cell. Binary cross-entropy (BCE) is used:

$$\mathcal{L}_{\text{obj}} = - [y_{\text{obj}} \log(p_{\text{obj}}) + (1 - y_{\text{obj}}) \log(1 - p_{\text{obj}})]. \quad (5.3)$$

3) Classification Loss For multi-class detection, YOLO uses BCE or cross-entropy over class logits:

$$\mathcal{L}_{\text{cls}} = - \sum_{c=1}^C y_c \log(p_c). \quad (5.4)$$

Total Loss Overall, the YOLO loss encourages accurate localization, correct objectness estimation, and reliable class prediction:

$$\boxed{\mathcal{L}_{\text{YOLO}} = \lambda_{\text{box}} (1 - \text{IoU}) + \lambda_{\text{obj}} \text{BCE}_{\text{obj}} + \lambda_{\text{cls}} \text{CE}_{\text{cls}}} \quad (5.5)$$

5.3 Experimental Setup

All experiments were conducted on an NVIDIA Tesla T4 GPU with 16 GB memory. Training used the PyTorch backend with the SGD optimizer and learning rate fixed at 0.001 for both training stages. Standard ExDark preprocessing (image resizing and YOLO-compatible augmentations) was applied. Batch sizes were chosen based on GPU memory limits, and mixed-precision training was used where possible to accelerate computation.

Results

This chapter presents the experimental results of the PENet-enhanced YOLO framework. We first visualize the Laplacian pyramid components generated by PENet, followed by quantitative evaluation using standard object detection metrics. Experiments were conducted on the ExDark dataset with YOLOv8n and YOLOv8m as baseline detectors.

6.1 Laplacian Pyramid Components Visualization

The Laplacian pyramid decomposes the input image into multi-scale frequency components. High-frequency layers capture edges and fine textures, whereas low-frequency components capture illumination and coarse structure. Figure 6.1 shows sample Laplacian components obtained from ExDark images after decomposition into four scales.

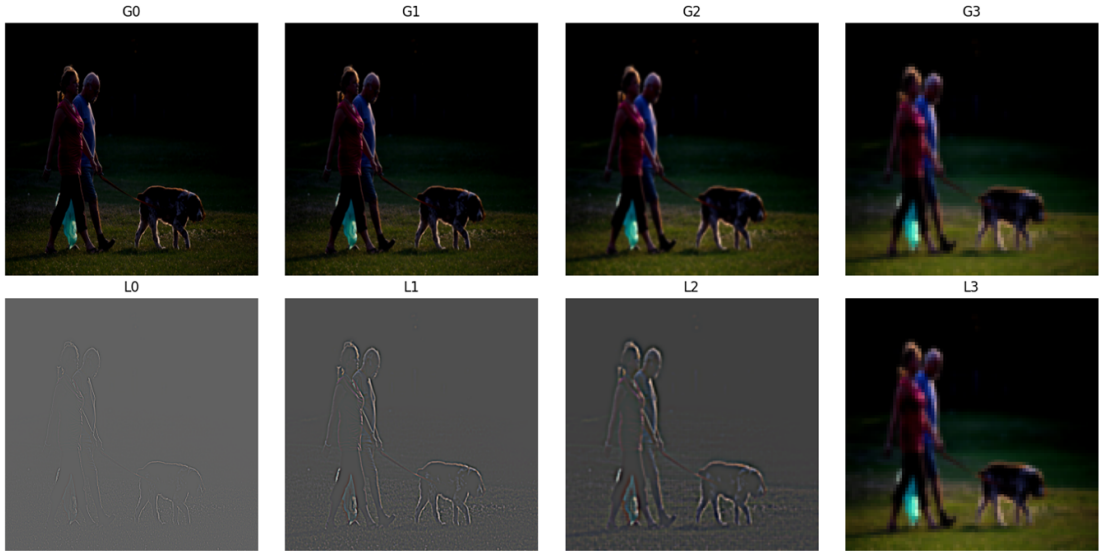


Figure 6.1: Visualization of Laplacian pyramid components used in PENet. L (High Freq) G (Low Freq) Each level contains high-frequency residuals highlighting edges and textures at progressively coarser scales.

These components serve as the input to the Detail Processing Module (DPM) and Low-Frequency Enhancement Filter (LEF), enabling PENet to separately process structural

details and illumination cues before reconstruction. **Input Resolution:**
All dataset images are resized to 640×640 before being passed into the model.

Gaussian Pyramid (Low-Frequency) Levels:

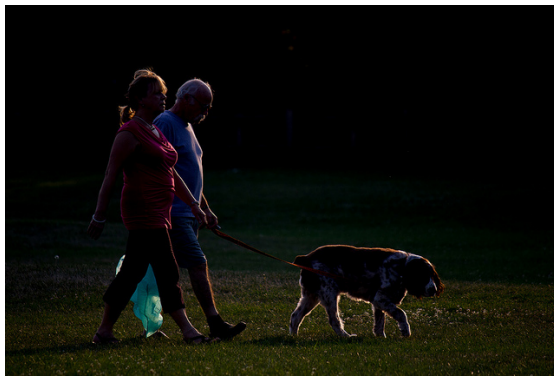
- **G0:** 640×640 (torch.Size([1, 3, 640, 640]))
- **G1:** 320×320 (torch.Size([1, 3, 320, 320]))
- **G2:** 160×160 (torch.Size([1, 3, 160, 160]))
- **G3:** 80×80 (torch.Size([1, 3, 80, 80]))

Laplacian Pyramid (High-Frequency) Levels:

- **L0:** 640×640 (torch.Size([1, 3, 640, 640]))
- **L1:** 320×320 (torch.Size([1, 3, 320, 320]))
- **L2:** 160×160 (torch.Size([1, 3, 160, 160]))
- **L3:** 80×80 (torch.Size([1, 3, 80, 80]))

6.1.1 How YOLO sees the Input image from PENet

Since we are passing the input image to the PENet and after components wise enhancement, Reconstructed Enhanced image is sent to the YOLO for detections. Figure ?? shows a comparison between the original low-light input and the enhanced output generated by PENet. YOLO processes only the enhanced representation, which acts as the true input to the backbone.



(a) Original Low-Light Input



(b) Enhanced Image from PENet

Figure 6.2: Comparison of the original input and the PENet-enhanced image fed into YOLO. The enhanced output exhibits restored edges, clearer object structure allowing YOLO to extract stronger features for detection.

6.2 Evaluation Metrics

Intersection over Union (IoU)

Intersection over Union measures the overlap between a predicted bounding box B_p and the ground truth box B_{gt} :

$$\text{IoU} = \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|}.$$

A prediction is considered correct if its IoU with the ground truth exceeds a chosen threshold.

Mean Average Precision (mAP)

Average Precision (AP) is computed as the area under the precision–recall curve for each class. The mean of AP across all object classes yields the mean Average Precision:

$$\text{mAP} = \frac{1}{C} \sum_{c=1}^C \text{AP}_c.$$

mAP@50

The term **mAP@50** refers to mAP computed using a fixed IoU threshold of 0.50, i.e., a prediction is counted as a true positive if $\text{IoU} > 0.5$. This is a commonly used metric in lightweight and real-time detection tasks due to its balanced tolerance between localization error and detection accuracy.

Precision and Recall

Precision measures the proportion of predicted detections that are correct, while recall measures the proportion of ground truth objects that are successfully detected:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

High precision indicates fewer false alarms, whereas high recall indicates fewer missed detections.

These metrics collectively evaluate how accurately the model localizes objects (IoU), how reliably it detects them (precision/recall), and how well it performs across all classes (mAP).

6.3 Model Evaluation

We evaluate the baseline YOLOv8 models and the proposed PENet-enhanced variants using mean Average Precision (mAP@50), recall, and the total number of trainable parameters. Results for both nano and medium configurations are summarized in Table 6.1.

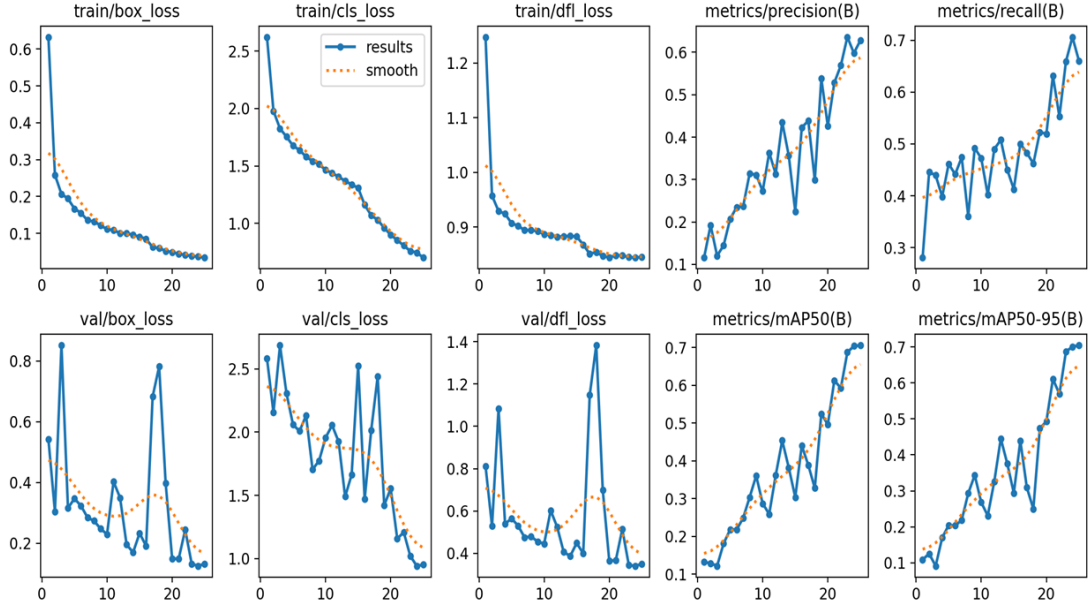


Figure 6.3: Training metrics of PENet + YOLOv8n

6.3.1 Quantitative Evaluation

The detection performance of the proposed enhancement-integrated models is evaluated using standard metrics such as Precision, Recall, and mAP@50. We compare the PENet + YOLO variants against their respective baseline models and further provide a detailed class-wise analysis to highlight category-level improvements.

Table 6.1: Performance comparison between baseline YOLOv8 and PENet-enhanced YOLO models on ExDark. (Metrics shown with placeholder values; will be updated after final training.)

Model	Params (M)	mAP@50%
YOLOv8n (Baseline)	3.0	65.70 [18]
PENet + YOLOv8n	3.09	73.50
YOLOv8m (Baseline)	25.0	79.60
PENet + YOLOv8m	25.09	85.00

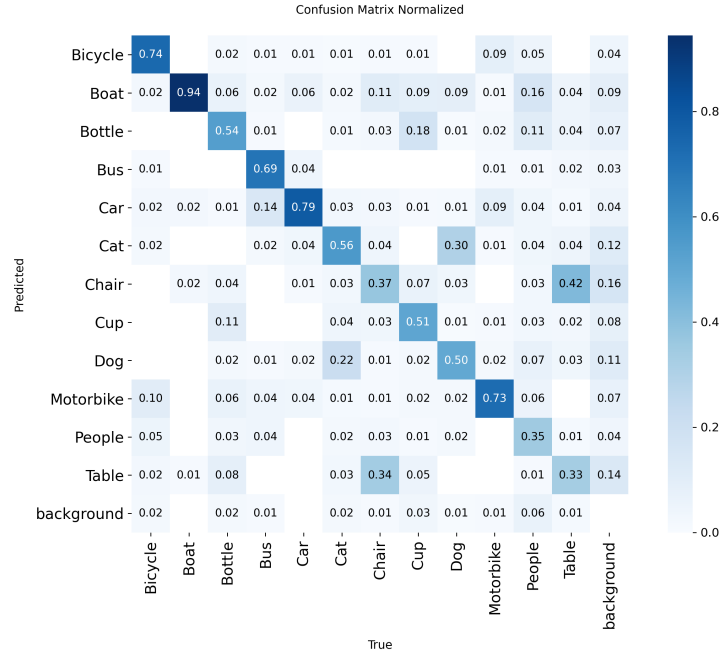


Figure 6.4: Confusion matrix of predictions of 12 classes

Table 6.2: Class-wise detection performance of PENet-enhanced YOLOv8n and YOLOv8m on the ExDark dataset.

Class	PENet + YOLOv8n			PENet + YOLOv8m		
	P	R	mAP50	P	R	mAP50
All	0.628	0.664	0.703	0.783	0.791	0.850
Bicycle	0.820	0.765	0.842	0.909	0.876	0.935
Boat	0.556	0.960	0.933	0.921	0.984	0.983
Bottle	0.610	0.589	0.650	0.834	0.741	0.854
Bus	0.841	0.758	0.878	0.892	0.918	0.960
Car	0.699	0.796	0.797	0.833	0.885	0.936
Cat	0.553	0.685	0.657	0.727	0.804	0.876
Chair	0.522	0.587	0.515	0.619	0.630	0.683
Cup	0.664	0.593	0.683	0.743	0.792	0.853
Dog	0.545	0.476	0.626	0.735	0.697	0.810
Motorbike	0.628	0.802	0.774	0.873	0.832	0.922
People	0.612	0.319	0.531	0.656	0.716	0.725
Table	0.482	0.633	0.553	0.652	0.611	0.660

6.3.2 Qualitative Evaluation

The qualitative results in Figure 6.5 show clear improvements achieved by the PENet-enhanced model under low-light conditions. In the first row, the baseline YOLOv8n fails to detect the boat instances on the right due to severe darkness, whereas the PENet+YOLO model recovers these objects, though the largest boat in the ground truth remains missed. In the second row, the baseline model does not detect the bus class and produces only high-confidence car predictions, while the enhanced model correctly identifies the bus and multiple vehicles. However, the confidence and consistency for the person class remain weak, indicating that the model still struggles with small and low-texture human regions in dark scenes.

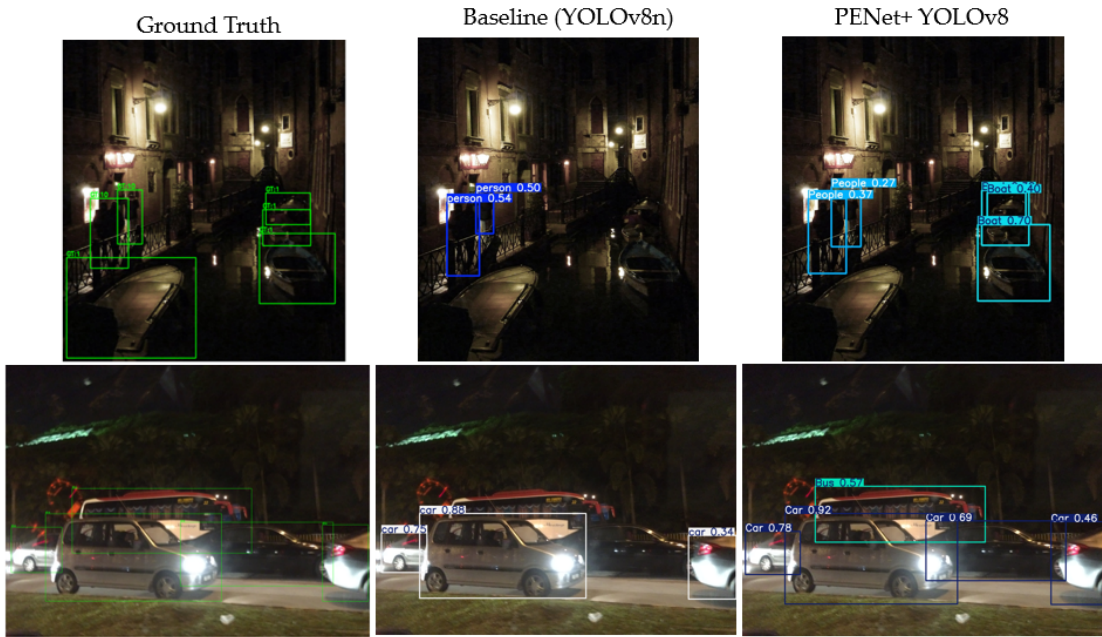


Figure 6.5: Detection Boxes from combined model on some samples

6.3.3 Failure Case Analysis

This section highlights failure cases observed under extreme low-light conditions.

(1) People Class Low mAP value

From Table 6.2, the people class shows noticeably weaker performance compared to other classes, with even the baseline model giving higher confidence scores. A likely reason is the over-enhancement introduced by the LEF module, as seen in Figure 6.2. This issue can be mitigated by using a weighted combination of high- and low-frequency components during reconstruction.

(2) People and Dogs in Extreme Darkness

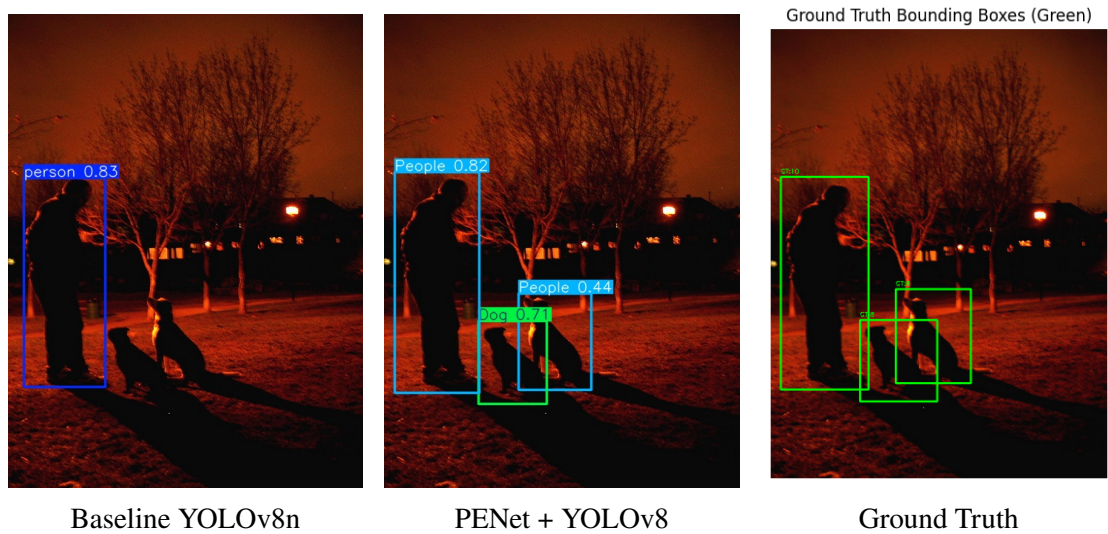


Figure 6.6: Failure Case 1: Missed dogs and weak person detection in extremely dark scenes.

- Baseline detects only one person and misses both dogs.
- PENet+YOLO detects 3 objects but classified dog as people
- Indicates difficulty when silhouettes merge with shadows.

(3) Object Misclassification

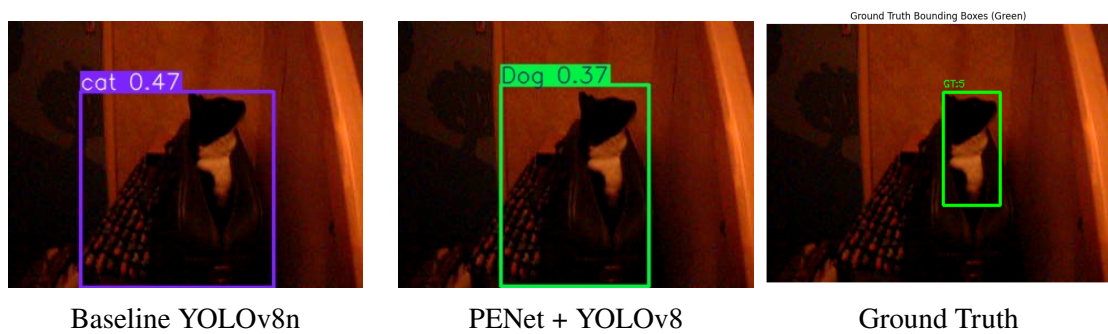


Figure 6.7: Failure Case 2: Cat incorrectly classified as Dog under texture loss.

- Baseline predicts cat with low confidence.
- PENet+YOLO misclassifies the same object as dog.
- Shows the model's sensitivity to texture loss in extremely dark settings.

(4) Missed Small or Distant Objects



Figure 6.8: Failure Case 3: Missed small/distant bicycles barely visible in deep shadows.

- Ground truth shows multiple bicycles including a small distant one.
- Baseline detects only the foreground bicycle; distant object remains invisible.
- PENet+YOLO enhances illumination locally but still fails to detect deep, low-contrast objects.

6.4 Discussion

The results demonstrate that integrating PENet with YOLO significantly improves detection performance in low-light environments. The Laplacian-based decomposition enables PENet to enhance structural and illumination cues before they reach the detector, leading to clear gains in both mAP and per-class precision–recall metrics. The medium variant (YOLOv8m + PENet) shows the highest improvement, confirming that stronger backbones benefit more from enhancement-driven feature refinement.

Class-wise analysis highlights consistent improvements across most categories, especially for large and well-structured objects such as **boat**, **bus**, and **motorbike**. However, certain classes—particularly **people** and small distant objects—remain challenging due to extreme darkness, loss of fine texture, or silhouettes merging with background shadows. Qualitative examples further show that while PENet boosts visibility, excessive enhancement may distort small object textures, leading to occasional misclassification.

Overall, the experiments validate the effectiveness of enhancement-guided detection, while also revealing areas for refinement, such as adaptive enhancement strength, dynamic frequency weighting, and ROI-aware enhancement for improved tracking stability.

Future Work and Conclusion

The present work Focused on foundation for enhancement-guided low-light object detection. Main aim of this work till now was to complete Optimized Detection model. In the current version, There are some improvements required on PENet architecture.

7.1 Enhancement Module Improvements

Future work will focus on refining PENet to improve its learning behaviour and representation capability. Two key modifications that can be made are:

- **Dynamic Sobel Filtering:** Instead of fixed horizontal and vertical Sobel kernels in the Edge Branch, learnable dynamic gradient filters can be introduced to adaptively capture edge variations under different illumination conditions. This allows the network to extract more discriminative high-frequency cues during training. The failure case-4 (Figure 6.8).
- **Adaptive Frequency Balancing for Stable Reconstruction:** As observed in the failure case (Figure 6.2), the low-frequency branch can over-amplify global illumination while the high-frequency branch contributes insufficient structural detail. Future work can introduce adaptive, resolution-aware frequency weighting across pyramid levels so that the network dynamically regulates the contribution of LF and HF components. This would prevent LF dominance, preserve fine textures, and produce more stable enhanced outputs for downstream detection.

7.2 Tracker Integration Through ROI Enhancement

Following the detector, the next phase of the system involves integrating a tracker that benefits from enhancement-guided feature refinement. Inspired by ROI-level enhancement strategies such as those used in DarkLighter [4], the plan is to enhance only the object regions required for tracking rather than the entire frame. This reduces computational overhead and sharpens tracking features, improving object identity preservation and reducing ID-switch occurrences in challenging low-light sequences.

7.3 End to End Detection–Tracking Pipeline

A complete detection–tracking system will be constructed by combining the enhanced detector with a lightweight tracker. The long-term goal is to design a unified logic that adaptively switches between detection and tracking based on confidence, motion cues, and temporal consistency. This will reduce redundant detection calls on video streams, increase runtime efficiency, and improve temporal coherence in low-light conditions.

7.4 Conclusion

In this work we proposed a complete system for Object Detection and tracking and focused on researching Detection models and particularly Pyramid Based Enhancement Framework (PENet) to improve low light image quality for detection tasks. By decomposing images into multi resolution frequency components and enhancing them separately, the combined model integrated with YOLOv8 aimed to correct illumination, preserve edges and textures. Experimental analysis showed certain failure cases like people class mAP was low, reason being over illumination (over smoothing) and loss of textures due to reconstruction method. Considering all these Observations, our future work focus on further enhancing the detection model, followed by tracker integration. Overall the project work establishes a foundation for frequency aware enhancement pipelines and opens clear pathways to further improve robustness for real world applications.

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